Motion Discrimination with Minimum Number of Channels Using Low-Cost Myoelectric Sensors in Myoelectric Prosthetic Hands

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***Abstract*—A myoelectric prosthetic hand has the possibility of improving the QoL of patients who have an upper-limb amputation. However, only 2.3% of patients use a myoelectric prosthetic hand in Japan[1]. One of the reason is the price. Most myoelectric prosthetic hand equips many myoelectric sensors and motors to estimate an intention and control hands accurately[2].**

In this paper, we discriminates a motion from electromyogram signal recorded by one reasonable myoelectric sensor as a basis of a low-cost and minimum channel control system. Electromyograms for five movements are measured on body surface near four major muscles in the forearm. Feature parameters extracted by electromyogram signals of one muscle is used to classify the motions. The discrimination is performed by feature extracted by one muscle and discriminator of a support vector machine and k-nearest neighbor. We show the discrimination performance of the proposed system using a single channel of the sensor. In addition, the sensor position which maximizes the discrimination performance of our proposal is identified.

I. Introduction

It is reported that there are more than 80,000 upper-limb amputees in Japan[3]. Upper limb amputees need prosthetic hands to realize a daily life similar to that of a non-handicapped person. Prosthetic hands are divided into three types: decorative prosthetic hands, active prosthetic hands, and myoelectric prosthetic hands. Myoelectric prosthetic hands have attracted attention for their appearance and function and for enabling finger and arm movements[4]. However, the penetration rate of myoelectric prosthetic hands in Japan is only 2.3%[1]. One of the causes is the high price of a single myoelectric prosthetic hand, which costs approximately 2 to 3 million yen[5]. In addition, a rehabilitation for operation of myoelectric prosthetic hands is only available in a limited number of facilities[6]. One of the reasons myoelectric prosthetics are so expensive is due to the use of several high-priced electrodes costing approximately 140,000 yen[2]. If it is possible to develop a myoelectric prosthetic hand with sufficient motion discrimination accuracy using low-cost electrodes or reducing the number of electrodes, the price of the myoelectric prosthetic hand can be reduced. This improvement yields the widespread use of myoelectric prosthetic hand.

As a study focusing on the number of electrodes, Kiso et al. verified the accuracy of motion discrimination by the minimum number of myoelectric channels. Two channels were placed on the side of the forearm to record electromyogram (EMG) signals. The results of neural network based discrimination of muscle potentials obtained by four movements of “open”, “grip”, “flexion”, and “extension” showed an average discrimination rate of more than 90% [7]. However, the use of two channels and the use of expensive electrode of the SX230 (120,000 yen)[8] were reported as problems. The EMG-based hand gesture estimation using continuous wavelet transform and long short-term memory - convolutional neural network (LSTM-CNN) was proposed by Nahid [9]. This paper discriminates the 10 class hand gestures 99% or more. However, the data set for the proposal includes a high time resolution and multi-channel data.

In this paper, we propose a five motions discrimination using one reasonable myoelectric sensor of MyoWare[10] to achieve a low-cost EMG-based motion identification. EMGs for five movements are measured on a body surface near four major muscles in the forearm. The target motions are: “neutral”, “open”, “grip”, “flexion”, and “extension”. The feature of EMG signals is classified into 5 motions using a support vector machine (SVM) and k-nearest neighbor (k-NN). In a training phase, parameters of SVM and k-NN, and measurement position is optimized to yield the maximum discrimination rate. From results, we discuss the discrimination rate of our proposal and whether it is possible to identify multiple motions with a single channel EMG.

II. Measurement Environment

The measurement schematic, the measurement conditions, and the measurement paradigm are shown in Fig. 1, Table I, and Fig. 2, respectively

The subject is seated on a stool with a natural posture. The subject’s forearm is stretched forward. When the measurement target is the flexor muscle, the arm is placed on a desk with the palm facing up (Fig. 1(a)). If the measurement is of the extensor muscle, the arm is placed with the palm facing down (Fig. 1(b)). This state is defined as a resting state without a motion.

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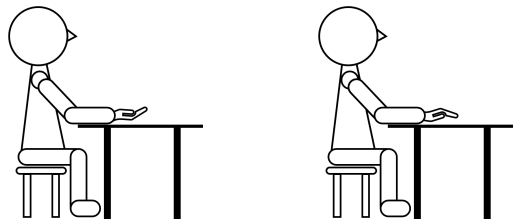

(a) Flexor (b) Extensor

Fig. 1: Measurement schematic

TABLE I: Measurement conditions

EMG	MyoWare
AD Converter	MCP3208
Output	Raspberry Pi 4 Model B
Sampling freq.	1,000Hz
Subject	3 adult males (normal persons)
Number of measurement	4 sets(400 data)
Position	Flexor digitorum superficialis, extensor digitorum, flexor carpi ulnaris, extensor carpi radialis longus
Motion	Neutral, open, grip, flexion, extension

Our purpose is to confirm whether it is possible to identify hand motions with a single channel by measuring the EMG potentials. The number of target motions is 5, its measurement positions is 4 (Table I). The position is a body surface near 4 major muscles in the forearm. There are 20 different measurements targeting 5 motions x 4 positions in our experiment. The EMG signal for single measurement is a data for 2 seconds of continuous motion. The subject performs the intended motion 5 times in 60 seconds at same interval (Fig. 2). The motion is standardized by performing the measurement with the instruction of "Please continue the motion for 2 seconds and squeeze your arm muscle as possible". A set of measurement consists of 100 EMG signals for 5 motions x 4 positions x 5 times. It is known that the frequency component of myopotential signal shifts to lower frequency band due to muscle fatigue[11]. One set of measurement is performed in a day to avoid the effect of muscle fatigue. We collected 400 signals per subject by taking four sets of measurements. The number of data for each measurement site per subject is 100 in this task.

III. Feature Parameter

Since the state of the muscles is constantly changing, the EMG signal measured on the skin surface is a non-stationary signal[12]. To represent characteristics of non-stationary signals, the mean, standard deviation, maximum of the integrated EMG (IEMG) and zero crossings obtained from the EMG signal are used as feature parameters of discriminator.

A. Preprocessing

Measured EMG signals used herein contain spike noises due to a transmission error of sensor and so on. To reduce these impulsive noises, a 5-point 1D moving median filter is applied as a preprocessing step.

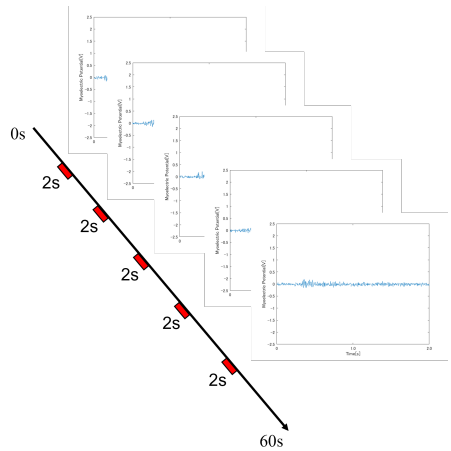

Fig. 2: Measurement paradigm

B. Integrated Electromyography

An IEMG is the amplitude information contained in a surface EMG measured using electrodes over the skin surface. An IEMG is obtained by applying Average Rectified Value (ARV) processing to the surface EMG[13]. The formula for calculating IEMG is shown in the equation (1).

$$IEMG(t) = \int_{-\infty}^{\infty} h(\tau)|x(t+\tau)|\, d\tau. \tag{1}$$

Where $x(t)$ is the 2 seconds of surface EMG at time t, and $h(\tau)$ is the weight function to calculate the integral. A MyoWare, the myopotential sensor in this study, is equipped with a pin that can apply ARV in the sensor and output IEMG. Therefore, an IEMG used as the features is obtained from the output of IEMG acquisition pin in MyoWare. The surface an EMG and IEMG for one target motion are shown in Fig. 3 and Fig. 4.

C. Zero Crossing

The zero crossing is simple frequency information and is the number of times the signal crosses the origin during the determined period. The formula of zero crossing is shown in the equation (2).

$$\begin{aligned} x(t) \cdot x(t+1) < 0 \quad AND \\ |x(t) - x(t+1)| > zc \\ (t = 1, 2, \ldots T_n - 1). \end{aligned} \tag{2}$$

In equation (2), $x(t)$ is the surface EMG at time t in the clipped period, and T_n is the length of the period. zc is the threshold derived by the equation (3) from a surface EMG at a neutral position.

$$zc = \mu + 3 \times \sigma. \tag{3}$$

μ is the mean value of the voltage in a surface EMG at a neutral, σ is the standard deviation. The length of the period (T_n) is 2000 which corresponds to 2 second EMG signal.

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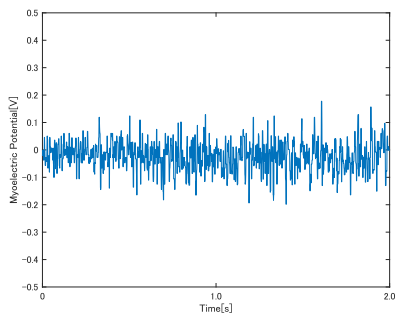

Fig. 3: Surface EMG

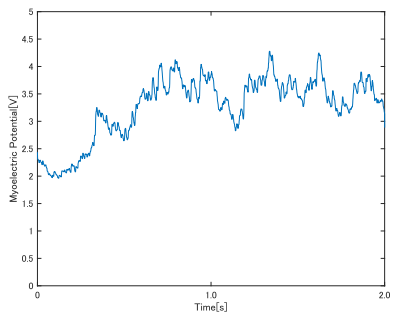

Fig. 4: IEMG

IV. Discriminator and Evaluation

SVM and k-NN are used as discriminators since the number of data is not enough to train a neural network based discriminator.

The motion discrimination problem in this study includes five motions. It is difficult to classify these motions using a linear separation classifier. Therefore, the kernel method is used for multi-class classification. The SVM in this research includes parameters of the cost C and the RBF kernel γ.

The parameters of SVM set to $C = 10$ and $\gamma = 1$. The parameter of k-NN is $k = 3$. These defined parameters yields the maximum classification rate for each discriminator.

5-fold stratified cross validation is used to evaluate a classification performance. The dataset is randomly divided into 5 parts. 1 of 5 parts is treated as the test data, and 4 parts are the training data. The number of training data is 80 and the number of test data is 20 to perform motion discrimination at each measurement position. Discrimination rates are calculated for each electrode location to show which position is better to classify five motions. The discrimination rate is defined as the equation (4).

$$Discrimination\ rate = \frac{Number\ of\ correct\ answers\ to\ test\ data}{Number\ of\ test\ data}. \quad (4)$$

V. Results

Motion discrimination rates for each measurement position are shown in Fig. 6 to Fig. 9. Fig. 6 to 8 are the rate for subjects A to C, and Fig. 9 is the average for all subjects. In each figure, the vertical axis is the discrimination rate and

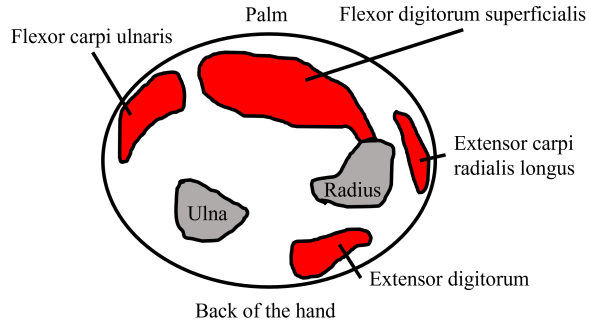

Fig. 5: Measurement positions

the horizontal axis is the measurement position. Measurement positions are, from left to right, flexor digitorum superficialis, extensor digitorum, flexor carpi ulnaris, and extensor carpi radialis longus.

From Fig. 6(a) and Fig. 8, the motion discrimination rate with the flexor digitorum superficialis is the highest for the SVM of subject A and the SVM and k-NN of subject C. From Fig. 6(b) and Fig. 7, the flexor carpi ulnaris is the highest discrimination rate for the other subjects and discriminators. In terms of the overall subject discrimination rate, Fig. 9 shows that the discrimination rate by the flexor digitorum superficialis and the flexor carpi ulnaris is higher than other muscles in the motion discrimination by SVM. On the other hand, the rate of the flexor carpi ulnaris is the highest by k-NN.

VI. Consideration

In Fig. 6(a) and Fig. 8(a),(b), the motion discrimination rate using EMG features of the flexor digitorum superficialis is the highest at 80%, 84%, and 79%, respectively. In Fig. 6(b) and Fig. 7(a),(b), the rate for the flexor carpi ulnaris is the highest at 77%, 82%, and 79%, respectively. 4 muscles are divided into 2 groups. One is flexor muscles such as the flexor digitorum superficialis and flexor carpi ulnaris. Another one is extensor muscles group such as the extensor digitorum and extensor carpi radialis longus. These indicate that the discrimination rate is higher for the flexor muscles group than for the extensor muscles group. In general, the muscle of the flexor group is larger than its extensor group in the forearm. This means that larger potentials induced by flexor muscles enhance the EMG feature to capture the characteristics of the motion.

From Fig. 9(a), the motion discrimination rate is the highest for the entire subject with the flexor digitorum superficialis and flexor carpi ulnaris for SVM. Discrimination rates for 2 muscles are 78% for both. In the case of k-NN, the motion discrimination rate is the highest with the flexor carpi ulnaris with a discrimination rate of 77%. Therefore, the EMG of the flexor digitorum superficialis and flexor carpi ulnaris are useful for SVM. In addition, the EMG of the flexor carpi ulnaris is useful for k-NN.

Based on these results, it is possible to discriminate multiple motions with a single channel using a low-cost EMG sensor at

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a discrimination rate of approximately 75% or higher by using the flexor digitorum superficialis and flexor carpi ulnaris.

VII. Conclusion

In this paper, the motion discrimination based on myoelectric potential measured by a single low-cost myoelectric sensor was verified to realize motion discrimination in myoelectric prosthetic hands at a lower cost and with a minimum number of channels.

The results showed the highest motion discrimination rates for each subject is higher than 77% in the measurement of EMG potentials by the flexor digitorum superficialis and flexor carpi ulnaris. This rate is about 10% lower than conventional research[7]. However, the cost of electrode is approximately 1/20.

Future issues include improving the discrimination rate. In addition, the number of discrimination target motions needs to be increased because there are fewer types of motions compared to daily arm movements.

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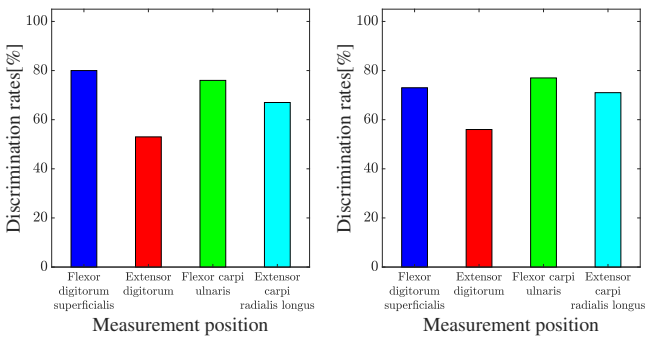
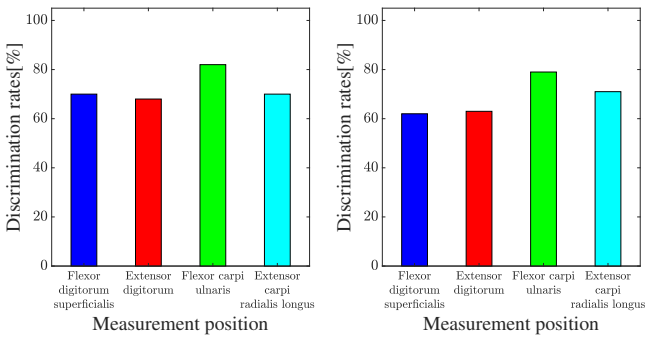


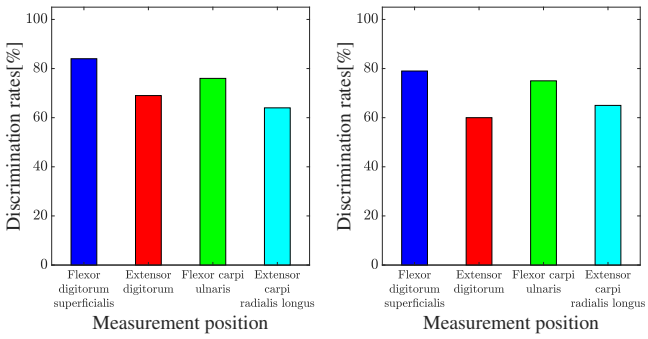
(a) SVM (b) k-NN

Fig. 6: Subject A



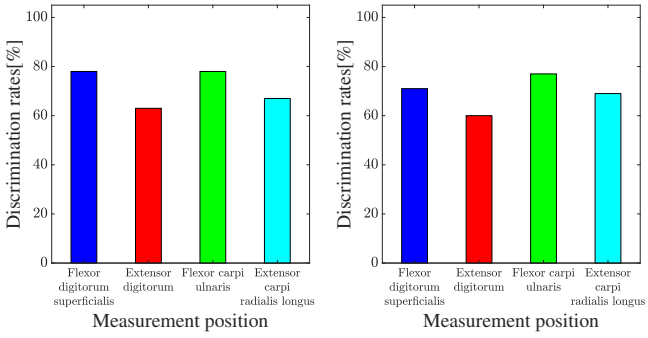
(a) SVM (b) k-NN

Fig. 7: Subject B



(a) SVM (b) k-NN

Fig. 8: Subject C



(a) SVM (b) k-NN

Fig. 9: All Subjects

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